

Missing data

Social and Cultural Analytics (7AAVBCS5)

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2026-10-19



Lecture Plan

- **Part I: Missing Data**
 - Understanding missing values
 - Detecting missing data
 - Handling strategies
 - Best practices
- **Part II: The cultural significance of missing data**
- **Part III: Changing a data frame**
 - Adding and removing columns
 - Adding and removing rows
 - Combining two tables
- **Part IV: Live Coding**

Key Learning Outcomes

By the end of this week you will be able to:

- Identify and handle missing data in Python
- Quantify how much is missing, and where

- Apply data cleaning techniques to real datasets
- Make informed decisions about missing data strategies
- Explain why missingness is itself evidence

Review: what we've learned so far

- Python as a calculator; types; variables
- Notebook basics
- Lists, indexing and slicing; numpy arrays
- Dictionaries
- Descriptive statistics: min, max, sum, mean, median, standard deviation
- f-strings
- Comparisons and `if/elif/else`
- Loading, selecting and filtering **data frames**
 - `pd.DataFrame()`, `.info()`, `.describe()`, `.loc`, `.iloc`, `.query()`
- Reading and saving CSV files
 - `pd.read_csv()`, `.to_csv(index=False)`

This week: the gaps in your data, and what they mean

Part I: Missing Data

Missing data

What is missing data?

- Data that should exist but doesn't
- Represented by `NaN` in pandas (“Not a Number”), or `None`
- pandas mostly treats `NaN` and `None` the same way

Why does data go missing?

- **Recording errors:** equipment failure, human error
- **Non-response:** survey participants skip questions
- **Impossible to record:** historical data, lost records
- **By design:** not applicable to certain cases

The issue:

- Can't be ignored
- Impacts statistical analyses

- Requires thoughtful handling strategies

Detecting missing values

```
1 import numpy as np
2 import pandas as pd
3
4 ages = pd.Series([23, 45, np.nan, 31])
5 print(ages.isna())
6 print("how many missing:", ages.isna().sum())
```

```
0    False
1    False
2     True
3    False
dtype: bool
how many missing: 1
```

Never test with `== None` or `== np.nan`. `np.nan == np.nan` is **False**. Always use `.isna()`.

Missing values break calculations – or quietly don't

```
1 ages = pd.Series([23, 45, np.nan, 31])
2 print("pandas sum: ", ages.sum())           # skips the gap
3 print("numpy sum:  ", np.array([23, 45, np.nan, 31]).sum())
```

```
pandas sum:  99.0
numpy sum:   nan
```

R's `sum()` returned `NA` unless you passed `na.rm = TRUE`.
pandas skips missing values by default — so nothing warns you that a number is based on fewer rows than you think.

How much is missing, and where

```
1 museum = pd.read_csv("../data/museum_data.csv")
2
3 missing_summary = pd.DataFrame({
4     "missing": museum.isna().sum(),
5     "missing_%": (museum.isna().mean() * 100).round(2),
6 })
7 missing_summary
```

	missing	missing_%
object_id	0	0.00
object_type	0	0.00
region	0	0.00
acquisition_date	0	0.00
acquisition_method	43	28.67
original_owner	134	89.33
cultural_significance	75	50.00
valuation	4	2.67

What to do when you find missing data

Ask:

1. **How much is missing?**
Quantify. Is it a little or a big problem?
2. **Why is it missing?** Random or systematic? If systematic, investigate the cause.
3. **What's your goal?**
Descriptive: document and report. Predictive: may need imputation.

Important:

- Always **document** your missing data and how you handled it
- Check if **missingness itself is informative**
- Consider creating a “missing” indicator variable
- Be transparent in reporting

Red flags:

- Never ignore gaps without investigating
- Avoid changing original data
 - create new columns

Types of Missingness

MCAR (Missing Completely At Random)

- Unrelated to any variable in your dataset
- *Random survey pages lost in the post*
- Safe to use any handling method

MAR (Missing At Random)

- Relates to other **observed** variables
- *Younger respondents skip income questions*
- Account for the pattern when handling

MNAR (Missing Not At Random)

- Relates to the **missing values themselves**
- *High earners refusing to report income*

- Most problematic — simple methods introduce bias
 - In cultural data:**
- Historical gaps often **MNAR** (power dynamics, deliberate omissions)
- Survey non-response often MAR
- Technical failures usually MCAR

Strategy 1: Deletion

```
1 print("rows we started with:", len(museum))
2 print("rows with no gaps at all:", len(museum.dropna()))
```

```
rows we started with: 150
rows with no gaps at all: 0
```

- **Pros:** simplest fix. Useful when only a few rows are affected
- **Cons:** loss of data, potential bias

On this dataset, deletion leaves **nothing**. Every one of the 150 objects is missing something.

Deletion, done deliberately

```
1 print("drop only rows with no valuation:", len(museum.dropna(subset=["valuation"])))
2 print("drop only columns that are half empty:",
3       museum.dropna(axis=1, thresh=int(0.5 * len(museum))).shape[1], "columns left")
```

```
drop only rows with no valuation: 146
drop only columns that are half empty: 7 columns left
```

Deletion is a scalpel, not a hammer: say **what** you are dropping and **why**.

Strategy 2: Imputation

```
1 filled = museum.copy()
2 filled["valuation"] = filled["valuation"].fillna(filled["valuation"].median())
3 print("missing before:", museum["valuation"].isna().sum())
4 print("missing after: ", filled["valuation"].isna().sum())
```

```
missing before: 4
missing after: 0
```

- **Pros:** keeps the rows that are real
- **Cons:** adds numbers that are **not** real — only estimates

Strategy 3: Keep and handle

```
1 print("mean of what is there:", round(museum["valuation"].mean(), 2))
2 print("...based on how many objects:", museum["valuation"].notna().sum(), "of", len(museum["valuation"]))
```

```
mean of what is there: 28198.63
...based on how many objects: 146 of 150
```

Some analyses tolerate gaps. **Report the denominator.**

Key points

- Missing values are `NaN` (or `None`) in pandas
- `.isna()` identifies them and returns True/False
- `.isna().sum()` counts them; `.isna().mean()` gives the proportion
- **pandas skips gaps by default** — so check before you trust a statistic
- Never compare to `NaN` with `==`
- Never modify the original: `.copy()` first, or make a new column

Part II: The cultural significance of missing data

Is the missingness patterned?

```
1 pd.crosstab(museum["acquisition_method"],  
2             museum["cultural_significance"].isna())
```

cultural_significance	False	True
acquisition_method		
Colonial Collection	5	24
Donation	12	10
Purchase	35	21

Counts hide it; percentages show it

```
1 (pd.crosstab(museum["acquisition_method"],  
2             museum["cultural_significance"].isna(),  
3             normalize="index") * 100).round(1)
```

cultural_significance	False	True
acquisition_method		
Colonial Collection	17.2	82.8
Donation	54.5	45.5
Purchase	62.5	37.5

82.8% of colonial-collection objects have no cultural significance recorded, against 37.5% of purchases.

Which kind of missingness is that?

The gap is **not** random. It relates to how the object was acquired — and plausibly to what the acquirer knew, or chose to record, or could not ask.

So: **MNAR**. Objects taken as colonial collections are the ones whose meaning went unrecorded.

Imputing a “most common” cultural significance here would invent exactly the knowledge that colonial acquisition destroyed.

Cultural implications

Missing data is a more important topic than simply how to handle it in Python.

Returning to weeks 1 and 2: the historical existence and chosen structure of data reflects **human decisions**.

When you come across missing data, ask:

1. What does it mean when certain cultural data is missing?
2. **Whose** data is more likely to be missing or excluded?
3. How does imputation affect interpretation in cultural contexts?

Hill, “Contextual Analytics”, §5.2 — what entered the dataset, what did not, and why

Part III: Changing a data frame

Adding and removing columns

```
1 work = museum.copy()
2
3 work["value_per_year"] = work["valuation"] / (2026 - work["acquisition_date"])
4 work["owner_recorded"] = work["original_owner"].notna()
5
6 work = work.drop(columns=["value_per_year"])
7 print(list(work.columns))
```

```
['object_id', 'object_type', 'region', 'acquisition_date', 'acquisition_method', 'original_owner',
'cultural_significance', 'valuation', 'owner_recorded']
```

Best practice, as in R: prefer a new data frame to modifying the original. Do not risk losing data.

Adding and removing rows

```
1 new_object = pd.DataFrame([{
2     "object_id": 999, "object_type": "Mask", "region": "West Africa",
3     "acquisition_date": 2024, "acquisition_method": "Donation",
4     "original_owner": "Known Community", "cultural_significance": "Ceremonial",
5     "valuation": 12000.0,
6 }])
7
8 longer = pd.concat([museum, new_object], ignore_index=True)
9 print(len(museum), "->", len(longer))
```

```
150 -> 151
```

Combining two tables

```
1 region_notes = pd.DataFrame({
2     "region": ["West Africa", "Southeast Asia", "South America",
3               "Pacific Islands", "North Africa"],
4     "continent": ["Africa", "Asia", "South America", "Oceania", "Africa"],
5 })
6
7 merged = museum.merge(region_notes, on="region", how="left")
8 print(merged[["region", "continent"]].drop_duplicates().to_string(index=False))
```

region	continent
Pacific Islands	Oceania
Southeast Asia	Asia
West Africa	Africa
North Africa	Africa
South America	South America

Part IV: Live Coding

Exercise 1: quantify the gaps

1. Build the missing-data summary table for `museum_data.csv`
2. Which column is worst? Which is nearly complete?
3. How many rows survive `dropna()`? Why so few?
4. Drop only the rows missing a `valuation`. How many go?

Exercise 2: is the missingness patterned?

1. Cross-tabulate `acquisition_method` against missing `cultural_significance`
2. Do it again as row percentages. What changes?
3. Try the same for `original_owner`
4. Decide: MCAR, MAR, or MNAR? Defend it

Exercise 3: your own project data

Bring the dataset you are considering for the report.

1. Load it
2. Build its missing-data summary
3. Find one column where the missingness looks **patterned**
4. Write two sentences for your data overview about what that pattern might mean

Before next week

1. Complete `w04_worksheet.ipynb` in the workshop
2. Keep working on your dataset choice — the data overview is due in **Week 8**
3. Read Hill, “Contextual Analytics”, §5.2

Note

The quiz in next week’s lecture covers **this week’s** material.

Extra



Speaker notes